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6.6 Answer Explanations: Data Sufficiency

The following discussion of Data Insights is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Data Insights. The particular questions in this chapter are generally representative of the kinds of Data Insights questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 1 to 38 — Difficulty: Easy

1. Each car at a certain dealership is either blue or white. What is the average (arithmetic mean) sticker price of all the cars at the dealership?

   1. Of all the cars at the dealership, $\frac{1}{3}$ are blue and have an average sticker price of $21,000.
   2. Of all the cars at the dealership, $\frac{2}{3}$ are white and have an average sticker price of $24,000.

   Algebra Statistics

   Let $\sum_b$ and $\sum_w$ be the sum of the sticker prices, respectively and in dollars, of the blue cars and the white cars at the dealership, and let n be the number of cars at the dealership. Determine the value of $\frac{\sum_b + \sum_w}{n}$.

   1. Given that there are $\frac{1}{3}n$ blue cars having an average sticker price of $21,000, it follows that $\sum_b = \left(\frac{1}{3}n\right)21{,}000 = 7{,}000n$. Therefore, $\frac{\sum_b + \sum_w}{n} = \frac{7{,}000n + \sum_w}{n} = 7{,}000 + \frac{\sum_w}{n}$, which can have more than one possible value by suitably varying $\sum_w$ and n; NOT sufficient.
   2. Given that there are $\frac{2}{3}n$ white cars having an average sticker price of $24,000, it follows that $\sum_w = \left(\frac{2}{3}n\right)24{,}000 = 16{,}000n$. Therefore, $\frac{\sum_b + \sum_w}{n} = \frac{\sum_b + 16{,}000n}{n} = \frac{\sum_b}{n} + 16{,}000$, which can have more than one possible value by suitably varying $\sum_b$ and n; NOT sufficient.

   Taking (1) and (2) together, $\frac{\sum_b + \sum_w}{n} = \frac{7{,}000n + 16{,}000n}{n} = \frac{23{,}000n}{n} = 23{,}000$.

   The correct answer is C; both statements together are sufficient.

2. A box contains only white balls and black balls. What is the probability that a ball selected at random from the box is white?

   1. There are 100 balls in the box.
   2. There are 40 black balls in the box.

   Arithmetic Probability

   Determine the probability of selecting a white ball from a box that contains only white and black balls.

   1. Given that there are 100 balls in the box, it is impossible to determine the probability of selecting a white ball because there is no information on the white/black split of the 100 balls in the box; NOT sufficient.
   2. Given that there are 40 black balls in the box, it is impossible to determine the probability of selecting a white ball because there is no indication of either the total number of balls in the box or the number of white balls; NOT sufficient.

   Taking (1) and (2) together, there are 100 balls in the box, 40 of which are black. It follows that the number of white balls is 100 − 40 = 60 and the probability of selecting a white ball is $\frac{60}{100} = \frac{3}{5}$.

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The correct answer is C; both statements together are sufficient.

3. A certain bag contains only red balls, blue balls, and green balls. What percent of all the balls in the bag are red?
   1. The ratio of the number of red balls to the number of blue balls in the bag is 1:3.
   2. There are 2 green balls in the bag.

Arithmetic Ratio and Proportion

Let R, B, and G be the number, respectively, of red balls, blue balls, and green balls in the bag. We are to determine R as a percent of $R+B+G$, or equivalently, we are to determine the value of $\frac{R}{R+B+G}$.

1. Given that $R{:}B = 1{:}3$, the value of $\frac{R}{R+B+G}$ could be $\frac{1}{6}$ (use $R=1, B=3, G=2$), and the value of $\frac{R}{R+B+G}$ could be $\frac{2}{10}$ (use $R=2, B=6, G=2$); NOT sufficient.
2. Given that $G=2$, the value of $\frac{R}{R+B+G}$ could be $\frac{1}{6}$ (use $R=1, B=3, G=2$), and the value of $\frac{R}{R+B+G}$ could be $\frac{2}{10}$ (use $R=2, B=6, G=2$); NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the value of $\frac{R}{R+B+G}$ because the two examples above satisfy both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

4. Each of the 120 students in a certain dormitory is either a junior or a senior. How many of the juniors have credit cards?
   1. $\frac{2}{3}$ of the 120 juniors and seniors have credit cards.
   2. The number of seniors who have credit cards is 20 more than the number of juniors who have credit cards.

Algebra First-Degree Equations

Determine the number of juniors who have credit cards among the 120 students in a certain junior/senior dormitory.

1. Given that $\frac{2}{3}$ of the 120 students have credit cards, it follows that 80 students have credit cards. There is no information regarding the number of juniors in this group of 80; NOT sufficient.
2. Given that the number of seniors with credit cards is 20 more than the number of juniors with credit cards, it is impossible to determine how many juniors have credit cards because no information is given about the junior/senior split nor about the have/do not have credit cards split of the 120 students; NOT sufficient.

Taking (1) and (2) together, 80 students have credit cards from (1) and the number of seniors with credit cards is 20 more than the number of juniors with credit cards from (2). Thus, $J+S=80$ or $J+(J+20)=80$, which can be solved for a unique value of J.

The correct answer is C; both statements together are sufficient.

5. If the average (arithmetic mean) cost per sweater for 3 pullover sweaters and 1 cardigan sweater was \$65, what was the cost of the cardigan sweater?
   1. The average cost per sweater for the 3 pullover sweaters was \$55.
   2. The most expensive of the 3 pullover sweaters cost \$30 more than the least expensive.

Algebra Statistics

Letting P represent the average cost, in dollars, of 1 pullover sweater and C, the cost, in dollars, of the cardigan, it is given that $\frac{3P+C}{4}=65$ or $3P+C=260$. Determine the value of C.

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It is given that $\frac{\quad}{4} = 65$ or $3P + C = 260$. Determine the value of C.

1. It is given that $P = 55$. Therefore, $3P = 3(55) = 165$ and $C = 260 - 165 = 95$; SUFFICIENT.
2. Given that the most expensive pullover sweater cost \$30 more than the least expensive, it is impossible to determine the value of C. For example, if the price of the most expensive pullover sweater was \$60, the price of the least expensive was \$30, and the price of the other pullover sweater was \$40, then the value of $C = 260 - 60 - 30 - 40 = 130$. But if the price of the most expensive pullover sweater was \$60, the price of the least expensive was \$30, and the price of the other pullover sweater was \$50, then the value of $C = 260 - 60 - 30 - 50 = 120$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

6. In each quarter of 1998, Company M earned more money than in the previous quarter. What was the range of Company M's quarterly earnings in 1998?

   1. In the 2nd and 3rd quarters of 1998, Company M earned \$4.0 million and \$4.6 million, respectively.
   2. In the 1st and 4th quarters of 1998, Company M earned \$3.8 million and \$4.9 million, respectively.

   Arithmetic Statistics

   We know that for each of the quarters in 1998, Company M earned more money than in the previous quarter. Is it possible to determine the range of the company's quarterly earnings in 1998?

   1. Although we are told the value of the earnings for the 2nd and 3rd quarters, Company M's 4th quarter earnings could, consistent with statement 1, be any amount that is greater than the 3rd-quarter earnings. Likewise, the company's 1st-quarter earnings could be any positive amount that is less than the company's 2nd-quarter earnings. The difference between these two values would be the range, and we see that it cannot be determined; NOT sufficient.
   2. We are given the earnings for the 1st and 4th quarters, and we already know that, from quarter to quarter, the earnings in 1998 have always increased. We can thus infer that Company M's earnings for the 2nd and 3rd quarters are less than the 4th-quarter earnings but greater than the 1st-quarter earnings. The difference between the greatest quarterly earnings and the least quarterly earnings for 1998 is thus the difference between the 4th-quarter earnings and the 1st-quarter earnings—the values \$4.9 million and \$3.8 million, respectively, that we have been given; SUFFICIENT.

   The correct answer is B; statement 2 alone is sufficient.

7. The range of the heights of a group of high school juniors and seniors is 20 centimeters. What is the average (arithmetic mean) of the height of the tallest senior in the group and the height of the shortest junior in the group?

   1. The average of the heights of the juniors in the group is 165 centimeters.
   2. The average of the heights of the seniors in the group is 179 centimeters.

   Arithmetic Statistics

   Determine the average of the height of the tallest senior and the height of the shortest junior.

   1. Given that the average of the heights of the juniors is 165 cm, it is not possible to determine the average of the height of the tallest senior and the height of the shortest junior. For example, the heights of the juniors could all be 165 cm and there could be three seniors with heights 176 cm, 176 cm, and 185 cm. In this case the range of all the heights is 185 − 165 = 20 cm, the average of the heights of the juniors is 165 cm, and the average of the height of the tallest senior and the height of the shortest junior is $\frac{185 + 165}{2} = 175$. Or the heights of the seniors could all be 179 cm and there could be three juniors with heights 159 cm, 168 cm, and 168 cm. In this case the range of all the heights is 179 – 159 = 20 cm, the average of the heights of the juniors is 165 cm, and the average of the height of the tallest senior and the height of the shortest junior is $\frac{179 + 159}{2} = 169$; NOT sufficient.
   2. Given that the average of the heights of the seniors is 179 cm, it is not possible to determine the average of the height of the tallest senior and the height of the shortest junior because, for each of the examples used in

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the height of the tallest senior and the height of the shortest junior because, for each of the examples used in (1) above, the average of the heights of the seniors is 179 cm; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the average of the height of the tallest senior and the height of the shortest junior because each of the examples used in (1) above satisfies both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

8. In a certain factory, hours worked by each employee in excess of 40 hours per week are overtime hours and are paid for at $1\frac{1}{2}$ times the employee's regular hourly pay rate. If an employee worked a total of 42 hours last week, how much was the employee's gross pay for the hours worked last week?

   1. The employee's gross pay for overtime hours worked last week was $30.
   2. The employee's gross pay for all hours worked last week was $30 more than for the previous week.

   Arithmetic Applied Problems

   If an employee's regular hourly rate was $\$R$ and the employee worked 42 hours last week, then the employee's gross pay for hours worked last week was $40R + 2(1.5R)$. Determine the value of $40R + 2(1.5R) = 43R$, or equivalently, the value of R.

   1. Given that the employee's gross pay for overtime hours worked last week was $30, it follows that $2(1.5R) = 30$ and $R = 10$; SUFFICIENT.
   2. Given that the employee's gross pay for all hours worked last week was $30 more than for the previous week, the value of R cannot be determined because nothing specific is known about the value of the employee's pay for all hours worked the previous week; NOT sufficient.

   The correct answer is A; statement 1 alone is sufficient.

9. Did Insurance Company K have more than $300 million in total net profits last year?

   1. Last year Company K paid out $0.95 in claims for every dollar of premiums collected.
   2. Last year Company K earned a total of $150 million in profits from the investment of accumulated surplus premiums from previous years.

   Arithmetic Applied Problems

   Letting R and E, respectively, represent the company's total revenue and total expenses last year, determine if $R - E > \$300$ million.

   1. This indicates that, for $\$x$ in premiums collected, the company paid $\$0.95x$ in claims, but gives no information about other sources of revenue or other types of expenses; NOT sufficient.
   2. This indicates that the company's profits from the investment of accumulated surplus premiums was $150 million last year, but gives no information about other sources of revenue or other types of expenses; NOT sufficient.

   Taking (1) and (2) together gives information on profit resulting from collecting premiums and paying claims as well as profit resulting from investments from accumulated surplus premiums but gives no indication whether there were other sources of revenue or other types of expenses.

   The correct answer is E; both statements together are still not sufficient.

10. How many hours would it take Pump A and Pump B working together, each at its own constant rate, to empty a tank that was initially full?

    1. Working alone at its constant rate, Pump A would empty the full tank in 4 hours 20 minutes.
    2. Working alone, Pump B would empty the full tank at its constant rate of 72 liters per minute.

    Arithmetic Applied Problems

    Determine how long it would take Pumps A and B working together, each at its own constant rate, to empty a full tank.

    1. This indicates how long it would take Pump A to empty the tank but gives no information about Pump B's constant rate; NOT sufficient.
    2. This indicates the rate at which Pump B can empty the tank, but without information about the capacity of

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2. This indicates the rate at which Pump B can empty the tank, but without information about the capacity of the tank or Pump A's rate, it is not possible to determine how long both pumps working together would take to empty the tank; NOT sufficient.

Taking (1) and (2) together gives the amount of time it would take Pump A to empty the tank and the rate at which Pump B can empty the tank, but without knowing the capacity of the tank, it is not possible to determine how long the pumps working together would take to empty the tank.

The correct answer is E; both statements together are still not sufficient.

11. Maria left home $\frac{1}{4}$ hour after her husband and drove over the same route as he had in order to overtake him. From the time she left, how many hours did it take Maria to overtake her husband?

    1. Maria drove 60 miles before overtaking her husband.
    2. While overtaking her husband, Maria drove at an average rate of 60 miles per hour, which was 12 miles per hour faster than her husband's average rate.

Arithmetic Rate Problems

1. Given that Maria drove 60 miles before overtaking her husband, it is not possible to determine how many hours she spent driving this distance. For example, she could have been driving this distance at a rate of 30 miles per hour and thus spent 2 hours driving this distance. However, she could also have been driving this distance at a rate of 60 miles per hour and thus spent 1 hour driving this distance; NOT sufficient.
2. Given that Maria drove at an average of 60 miles per hour and her husband drove at an average of 60 – 12 = 48 miles per hour, and letting t be the number of hours it took for Maria to overtake her husband, it follows that $60t = 48(t + \frac{1}{4})$ since the distance Maria drove, $60t$ miles, is the same as the distance her husband drove, $48(t + \frac{1}{4})$ miles. Therefore, $60t = 48t + 12$, or $t = 1$, and hence it took 1 hour for Maria to overtake her husband; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

12. How many people received a certain survey?

    1. Six-tenths of those who received the survey responded.
    2. Of those who received the survey, 42 responded.

Arithmetic Equations

Let n be the number of people who received the survey. We are to determine the value of n.

1. Given only that the number of people who both received the survey and responded to the survey is $\frac{6}{10}n$, it is not possible to determine the value of n since any positive integer multiple of 5 (e.g., 5, 10, 15, 20, etc.) could be the value of n; NOT sufficient.
2. Given only that the number of people who both received the survey and responded to the survey is 42, it is not possible to determine the value of n since any integer greater than or equal to 42 could be the value of n; NOT sufficient.

Taking (1) and (2) together, it follows that $\frac{6}{10}n = 42$, which has exactly one solution for the value of n, namely $n = 70$.

The correct answer is C; both statements together are sufficient.

13. From 1985 to 1994, what was the percent increase in total United States trade?

    1. Total United States trade in 1985 was 17 percent of gross domestic product in 1985.
    2. Total United States trade in 1994 was 23 percent of gross domestic product in 1994.

Arithmetic Percents

1. Given that in 1985, the total United States trade was 17 percent of gross domestic product, it is not possible to determine the percent increase in total United States trade from 1985 to 1994 because no information

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to determine the percent increase in total United States trade from 1985 to 1994 because no information pertaining to 1994 is provided; NOT sufficient.

2. Given that in 1994, the total United States trade was 23 percent of gross domestic product, it is not possible to determine the percent increase in total United States trade from 1985 to 1994 because no information pertaining to 1985 is provided; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the percent increase in total United States trade from 1985 to 1994 because no information is provided or can be deduced about the ratio of the 1985 gross domestic product to the 1994 gross domestic product. For example, let N be the 1985 gross domestic product. If the 1994 gross domestic product were also N, then the percent increase in total trade from 1985 to 1994 would be $\frac{23N - 17N}{17N} \approx 35\%$. However, if the 1994 gross domestic product were $2N$, then the percent increase in total trade from 1985 to 1994 would be $\frac{23(2N) - 17N}{17N} \approx 171\%$.

The correct answer is E; both statements together are still not sufficient.

14. Company X's profits this year increased by 25% over last year's profits. Was the dollar amount of Company X's profits this year greater than the dollar amount of Company Y's?

1. Last year, the ratio of Company Y's profits to Company X's profits was 5:2.
2. Company Y experienced a 40% drop in profits from last year to this year.

Algebra Applied Problems

Let P_X and P'_X, respectively, be the profits of Company X last year and this year, and let P_Y and P'_Y, respectively, be the profits of Company Y last year and this year. Then $P'_X = 1.25P_X$. Is $P'_X > P'_Y$?

1. Given that $\frac{P_Y}{P_X} = \frac{5}{2}$, it is not possible to determine whether $P'_X > P'_Y$ because nothing is known about the value of P'_Y other than P'_Y is positive; NOT sufficient.
2. Given that $P'_Y = 0.6P_Y$, it is not possible to determine whether $P'_X > P'_Y$ because nothing is known that relates the profits of Company X for either year to the profits of Company Y for either year; NOT sufficient.

Taking (1) and (2) together, it is given that $P'_X = 1.25P_X$ and from (1) it follows that $\frac{P_Y}{P_X} = \frac{5}{2}$, or $P_X = \frac{2}{5}P_Y$, and thus $P'_X = (1.25)\left(\frac{2}{5}P_Y\right)$. From (2) it follows that $P'_Y = 0.6P_Y$, or $P_Y = \frac{1}{0.6}P'_Y$, and thus $P'_X = (1.25)\left(\frac{2}{5}\right)\left(\frac{1}{0.6}P'_Y\right)$. Since the last equation expresses P'_X as a specific number times P'_Y, it follows that it can be determined whether or not $P'_X > P'_Y$. Note that $(1.25)\left(\frac{2}{5}\right)\left(\frac{1}{0.6}\right) = \left(\frac{5}{4}\right)\left(\frac{2}{5}\right)\left(\frac{5}{3}\right) = \frac{5}{6}$, and so the answer to the question "Is $P'_X > P'_Y$" is no.

The correct answer is C; both statements together are sufficient.

15. A certain company consists of three divisions, A, B, and C. Of the employees in the three divisions, the employees in Division C have the greatest average (arithmetic mean) annual salary. Is the average annual salary of the employees in the three divisions combined less than $55,000?

1. The average annual salary of the employees in Divisions A and B combined is $45,000.
2. The average annual salary of the employees in Division C is $55,000.

Algebra Statistics

1. Given that the average annual salary of the employees in Divisions A and B combined is $45,000, each of the divisions could have exactly two employees such that the annual salaries in Division A are $45,000 and $45,000, the annual salaries in Division B are $45,000 and $45,000, and the annual salaries in Division C are $50,000 and $50,000, in which case Division C has the greatest average annual salary and the average

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annual salary in Divisions A, B, and C combined is less than \$55,000. On the other hand, each of the divisions could have exactly two employees such that the annual salaries in Division A are \$45,000 and \$45,000, the annual salaries in Division B are \$45,000 and \$45,000, and the annual salaries in Division C are \$1 million and \$1 million, in which case Division C has the greatest average annual salary and the average annual salary in Divisions A, B, and C combined is greater than \$55,000; NOT sufficient.

2. Given that the average annual salary in Division C is \$55,000, we have $\frac{\sum_C}{N_C} = 55{,}000$, where $\sum_C$ is the sum of the annual salaries, in dollars, of the employees in Division C and N_C is the number of employees in Division C. Moreover, letting $\sum_A$ and $\sum_B$ be the sums of the annual salaries, respectively and in dollars, of the employees in Divisions A and B, and letting N_A and N_B be the numbers of employees, respectively, in Divisions A and B, then we have $\frac{\sum_A}{N_A} < 55{,}000$ and $\frac{\sum_B}{N_B} < 55{,}000$, since the employees in Division C have the greatest average annual salary. Note that these two inequalities and this equation can be rewritten as $\sum_A < 55{,}000N_A$, $\sum_B < 55{,}000N_B$, and $\sum_C = 55{,}000N_C$. Therefore, the average annual salary of the employees in the three divisions combined is
$\frac{\sum_A + \sum_B + \sum_C}{N_A + N_B + N_C} = \frac{\sum_A + \sum_B + 55{,}000N_C}{N_A + N_B + N_C}$, which is less than
$\frac{55{,}000N_A + 55{,}000N_B + 55{,}000N_C}{N_A + N_B + N_C} = \frac{55{,}000(N_A + N_B + N_C)}{N_A + N_B + N_C} = 55{,}000$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

16. A candle company determines that, for a certain specialty candle, the supply function is $p = m_1x + b_1$ and the demand function is $p = m_2x + b_2$, where p is the price of each candle, x is the number of candles supplied or demanded, and m_1, m_2, b_1, and b_2 are constants. At what value of x do the graphs of the supply function and demand function intersect?

    1. $m_1 = -m_2 = 0.005$
    2. $b_2 - b_1 = 6$

    Algebra First-Degree Equations

    The graphs will intersect at the value of x such that $m_1x + b_1 = m_2x + b_2$ or $(m_1 - m_2)x = b_2 - b_1$.

    1. This indicates that $m_1 = -m_2 = 0.005$. It follows that $m_1 - m_2 = 0.01$, and so $0.01x = b_2 - b_1$ or $x = 100(b_2 - b_1)$, which can vary as the values of b_2 and b_1 vary; NOT sufficient.
    2. This indicates that $b_2 - b_1 = 6$. It follows that $(m_1 - m_2)x = 6$. This implies that $m_1 \neq m_2$, and so $x = \frac{b_2 - b_1}{m_1 - m_2} = \frac{6}{m_1 - m_2}$, which can vary as the values of m_1 and m_2 vary; NOT sufficient.

    Taking (1) and (2) together, $m_1 - m_2 = 0.01$ and $b_2 - b_1 = 6$ and so the value of x is $\frac{6}{0.01} = 600$.

    The correct answer is C; both statements together are sufficient.

17. A certain ski shop sold 125 pairs of skis and 100 pairs of ski boots for a total of \$75,000. What was the average (arithmetic mean) selling price of a pair of the ski boots?

    1. The average selling price of a pair of skis was \$300.
    2. The selling price of a pair of ski boots varied from \$150 to \$900.

    Arithmetic Statistics

    Let $\sum_{\text{skis}}$ be the sum of the selling prices, in dollars, of all 125 pairs of skis and let $\sum_{\text{boots}}$ be the sum of the selling prices, in dollars, of all 100 pairs of ski boots. We are given that $\sum_{\text{skis}} + \sum_{\text{boots}} = 75{,}000$. Determine the value of $\frac{\sum_{\text{boots}}}{100}$, or equivalently, determine the value of $\sum_{\text{boots}}$.

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1. Given that $\frac{\sum_{\text{skis}}}{125} = 300$, or $\sum_{\text{skis}} = 300(125) = 37{,}500$, it follows from $\sum_{\text{skis}} + \sum_{\text{boots}} = 75{,}000$ that $\sum_{\text{boots}} = 75{,}000 - \sum_{\text{skis}} = 75{,}000 - 37{,}500 = 37{,}500$; SUFFICIENT.
2. Given that the selling price of a pair of ski boots varied from \$150 to \$900, it is possible that there were 40 pairs of ski boots each with a selling price of \$150, 60 pairs of ski boots each with a selling price of \$900, and 125 pairs of skis each with a selling price of \$120 for a total selling price of 40(\$150) + 60(\$900) + 125(\$120) = \$75,000, and thus it is possible that $\sum_{\text{boots}} = 40(150) + 60(900) = 6{,}000 + 54{,}000 = 60{,}000$. However, it is also possible that there were 60 pairs of ski boots each with a selling price of \$150, 40 pairs of ski boots each with a selling price of \$900, and 125 pairs of skis each with a selling price of \$240 for a total selling price of 60(\$150) + 40(\$900) + 125(\$240) = \$75,000, and thus it is also possible that $\sum_{\text{boots}} = 60(150) + 40(900) = 9{,}000 + 36{,}000 = 45{,}000$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

18. Last year Publisher X published 1,100 books, consisting of first editions, revised editions, and reprints. How many first editions did Publisher X publish last year?
    1. The number of first editions published was 50 more than twice the number of reprints published.
    2. The number of revised editions published was half the number of reprints published.

Algebra Simultaneous Equations

Let A be the number of first editions, B be the number of revised editions, and C be the number of reprints. Then $A + B + C = 1{,}100$. Determine the value of A.

1. Given that $A = 50 + 2C$, it is not possible to determine the value of A. This is because by choosing different values of C, different values of A can be obtained by using the equation $A = 50 + 2C$, and then the equation $A + B + C = 1{,}100$ can be used to determine whether acceptable values of B (nonnegative integers) exist for these values of A and C. For example, choosing $C = 100$ leads to $A = 250$ and $B = 750$, and choosing $C = 200$ leads to $A = 450$ and $B = 450$; NOT sufficient.
2. Given that $B = \frac{1}{2}C$, or $C = 2B$, it is not possible to determine the value of A. This is because by choosing different values of B, different values of C can be obtained by using the equation $C = 2B$, and then the equation $A + B + C = 1{,}100$ can be used to determine different values of A. For example, choosing $B = 100$ leads to $C = 200$ and $A = 800$, and choosing $B = 200$ leads to $C = 400$ and $A = 500$; NOT sufficient.

Taking $A = 50 + 2C$ from (1) and $C = 2B$ from (2) together gives $A = 50 + 4B$. Thus, in the equation $A + B + C = 1{,}100$, A can be replaced with $50 + 4B$ and C can be replaced with $2B$ to give $(50 + 4B) + B + 2B = 1{,}100$. Solving for B gives $B = 150$, and hence $C = 2B = 300$ and $A = 50 + 2C = 650$.

The correct answer is C; both statements together are sufficient.

19. How old is Jane?
    1. Ten years ago she was one-third as old as she is now.
    2. In 15 years, she will be twice as old as she is now.

Algebra First-Degree Equations

Determine the value of J, where J represents Jane's current age.

1. In symbols, $J - 10$ represents Jane's age ten years ago and $\frac{1}{3}J$ represents one-third her current age. These expressions are equal by (1), so $J - 10 = \frac{1}{3}J$. This is a first-degree equation in the variable J and has a unique solution; SUFFICIENT.
2. In symbols, $J + 15$ represents Jane's age 15 years from now and $2J$ represents twice her current age. These expressions are equal by (2), so $J + 15 = 2J$. This is a first-degree equation in the variable J and

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These expressions are equal by (2), so $J + 15 = 2J$. This is a first-degree equation in the variable J and has a unique solution; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

20. What was the population of City X in 2002?

    1. X's population in 2002 increased by 2 percent, or 20,000 people, over 2001.
    2. In 2001, X's population was 1,000,000.

Algebra Percents

Letting P_1 and P_2 represent City X's population in 2001 and 2002, respectively, the percent increase in population from 2001 to 2002 is given as a decimal by $\frac{P_2 - P_1}{P_1}$.

1. By (1) the percent increase was 2 percent, so $\frac{P_2 - P_1}{P_1} = 0.02$ or $P_2 - P_1 = 0.02P_1$. Also, by (1), $P_2 - P_1 = 20{,}000$ so $20{,}000 = 0.02P_1$ from which the value of P_1 can be uniquely determined. Then $P_1 + 20{,}000 = P_2$, which is the population of City X in 2002; SUFFICIENT.
2. Even though (2) gives $P_1 = 1{,}000{,}000$, it gives no information about the population of City X in 2002 either by itself or in relation to the population in 2001; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

21. Yesterday Bookstore B sold twice as many softcover books as hardcover books. Was Bookstore B's revenue from the sale of softcover books yesterday greater than its revenue from the sale of hardcover books yesterday?

    1. The average (arithmetic mean) price of the hardcover books sold at the store yesterday was $10 more than the average price of the softcover books sold at the store yesterday.
    2. The average price of the softcover and hardcover books sold at the store yesterday was greater than $14.

Arithmetic Statistics

Letting s represent the number of softcover books sold; h, the number of hardcover books sold; S, the average price of the softcover books sold; and H, the average price of the hardcover books sold, determine whether the revenue from the sale of softcover books is greater than the revenue from the sale of hardcover books or if $sS > hH$, where $s = 2h$.

1. Given that $H = S + 10$, if $S = 10$, $H = 20$, $s = 10$, and $h = 5$, then $sS = 100$ and $hH = 100$, so $sS = hH$. On the other hand, if $S = 40$, $H = 50$, $s = 8$, and $h = 4$, then $sS = 320$ and $hH = 200$, so $sS > hH$; NOT sufficient.
2. Given that $\frac{sS + hH}{s + h} > 14$, if $s = 6$, $S = 10$, $h = 3$, and $H = 30$, $\frac{6(10) + 3(30)}{6 + 3} = \frac{150}{9} > 14$ and $6(10) < 3(30)$. On the other hand, if $s = 10$, $S = 15$, $h = 5$, and $H = 20$, $\frac{10(15) + 5(20)}{10 + 5} = \frac{250}{15} > 14$ and $10(15) > 5(20)$; NOT sufficient.

Taking (1) and (2) together,

$\frac{sS + hH}{s + h}$	>	14	from (2)
$\frac{2h(H - 10) + hH}{2h + h}$	>	14	$s = 2h$ (given) and $H = S + 10$ from (1)
$\frac{3H - 20}{3}$	>	14	cancel h and simplify
$3H - 20$	>	42	multiply both sides by 3

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H	$>$	$\frac{62}{3}$	solve for H

To show that this leads to $sS > hH$, start with $sS > hH$ and then reverse the steps.

sS	$>$	hH	
$2h(H-10)$	$>$	hH	$s = 2h$ and $S = H - 10$
$2hH - 20h$	$>$	hH	distributive property
$2hH$	$>$	$hH + 20h$	add $20h$ to both sides
hH	$>$	$20h$	subtract hH from both sides
H	$>$	20	divide both sides by $h > 0$

Now reverse the steps.

H	$>$	$\frac{62}{3}$	derived earlier
H	$>$	20	$\frac{62}{3} > 20$
hH	$>$	$20h$	multiply both sides by $h > 0$
$2hH$	$>$	$hH + 20h$	add hH to both sides
$2hH - 20h$	$>$	hH	subtract $20h$ from both sides
$2h(H-10)$	$>$	hH	factor
sS	$>$	hH	$s = 2h$ and $S = H - 10$

Thus, the revenue from the sale of softcover books was greater than the revenue from the sale of hardcover books.

The correct answer is C; both statements together are sufficient.

22. At a certain refreshment stand, all hot dogs are the same price, and all sodas are the same price. What is the total price of 3 hot dogs and 2 sodas at the refreshment stand?

    1. The total price of 5 sodas at the stand is less than the total price of 2 hot dogs.
    2. The total price of 9 hot dogs and 6 sodas at the stand is $21.

Algebra Applied Problems

Let $\$H$ be the price of each hot dog and $\$S$ be the price of each soda. We are to determine the value of $3H + 2S$.

1. Given that $5S < 2H$, the value of $3H + 2S$ could be 7.5 (if $H = 2$ and $S = 0.75$), and the value of $3H + 2S$ could be 11 (if $H = 3$ and $S = 1$); NOT sufficient.
2. Given that $9H + 6S = 21$ and recognizing that the left-hand side of this equation is 3 times $3H + 2S$

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2. Given that $9H + 6S = 21$ and recognizing that the left-hand side of this equation is 3 times $3H + 2S$ (what we want the value of), divide both sides of this equation by 3 to get $3H + 2S = 7$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

23. If a certain school will buy computers at a fixed price of x dollars each and computer desks at a fixed price of y dollars each, how many dollars will the school pay for each computer?

    1. The school will pay a total of $2,200 if it buys 2 computers and 1 computer desk at these fixed prices.
    2. The school will pay a total of $5,400 if it buys 5 computers and 2 computer desks at these fixed prices.

Algebra Simultaneous Equations

1. Given that $2x + y = 2{,}200$, by choosing different values for y and solving for x, it follows that more than one value of x is possible, and hence the value of x cannot be determined. For example, $x = 1{,}000$ is possible (choose $y = 200$), and $x = 900$ is possible (choose $y = 400$); NOT sufficient.
2. Given that $5x + 2y = 5{,}400$, by choosing different values for y and solving for x, it follows that more than one value of x is possible, and hence the value of x cannot be determined. For example, $x = 1{,}000$ is possible (choose $y = 200$), and $x = 920$ is possible (choose $y = 400$); NOT sufficient.

Taking (1) and (2) together, we have $2x + y = 2{,}200$ and $5x + 2y = 5{,}400$. These two equations can be solved simultaneously to obtain a unique value of x. This can be done by solving for y in terms of x in the first equation to get

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